Artificial Intelligence in Pediatric Radiology

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Objectives

- Describe the big players in radiology AI, both general and pediatric, since 2023
- Summarize what radiology AI does well and what remains unresolved

Methods

Academic output. PubMed (E-utilities) yearly counts per query and per modality/task term group, plus **preprints** (OpenAlex `type:preprint`: arXiv, medRxiv, bioRxiv, ...) counted with the same queries, because PubMed does not index arXiv (4,426 radiology-AI preprints in 2025). **Most-cited papers** from OpenAlex (union of modality/task searches, deduped, articles and preprints): one list for 2008-2022, then one per year from 2023, each showing raw citations and the field-weighted citation impact (FWCI, 1 = world average for that field and year), so recent papers are not buried under 2023 ones. **Venues**: Semantic Scholar venue search (NeurIPS, ICLR, ICML, CVPR, MICCAI, MIDL) and OpenAlex journal filters (RSNA, SPR), 2023–present; **PatentsView** for granted patents.

Clinical problems. The pediatric radiology-AI query AND a term group per problem (bone age, fracture, pneumonia, appendicitis, brain tumor, ...), counted per era; a paper can name several problems.

Datasets. Public pediatric imaging datasets, sizes verified against the primary papers or hosting pages.

Commercial players. The **FDA AI-enabled device list** (public spreadsheet: decision date, device, company, lead panel), filtered to the Radiology panel; cross-checked against a curated list of products with pediatric indications.

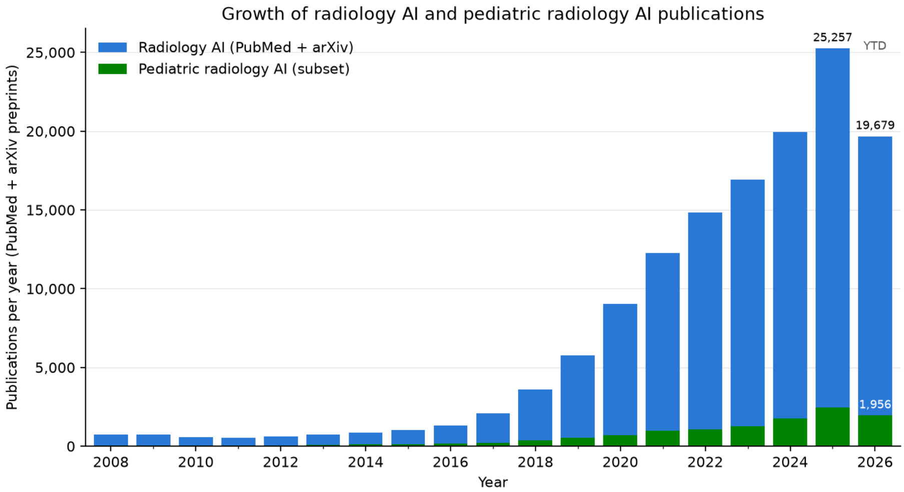
Trade press. Newsletter archives (The Imaging Wire, RSNA News, ESR/ECR, TLDR, Signify Research, Radiology Business) split into stories; a story is radiology-AI when AI terms co-occur with imaging terms, pediatric when pediatric terms also co-occur; a story's outbound publisher link is resolved to a DOI and cited (OpenAlex).

Pediatric filter. PubMed: the radiology-AI query AND (pediatric* OR paediatric* OR child* OR infant* OR neonat* OR adolescen* OR "children's hospital"). Most-cited lists: the title must also carry a pediatric term (bone age, fetal, newborn, ...), otherwise highly cited adult papers float in. Newsletters: pediatric term inside the same story.

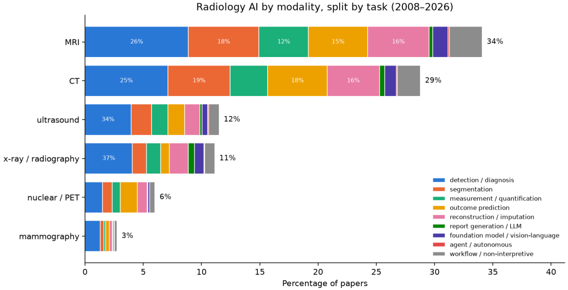
Representative PubMed query (radiology $\cap$ AI):

(radiology OR radiograph* OR "medical imaging" OR MRI OR "computed tomography" OR CT[tiab] OR ultrasound[tiab] ... NOT ("optical coherence" OR fundus OR dental OR histopatholog* ...)) AND ("artificial intelligence" OR "deep learning" OR "convolutional neural network" OR radiomics ...)

Radiology AI publication counts



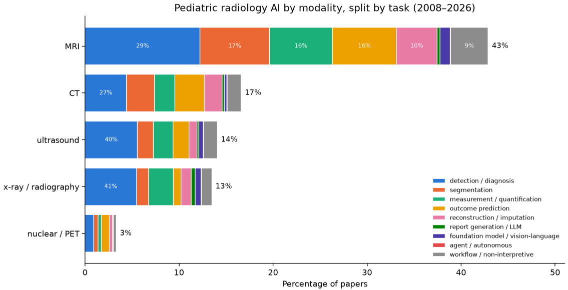
Radiology AI publications, stratified by modality and task



Bar = percentage of radiology-AI papers (123,413 = 98,842 PubMed + 24,571 preprints, 2008–present) whose title/abstract names the modality; segments = which task terms those papers use (overlapping). What each task means, by what the model produces:

- **detection / diagnosis**: is a finding present now, and which one (label or box): fracture, pneumonia, tumor; includes triage and screening
- **segmentation**: outline a structure or lesion (a mask)
- **measurement / quantification**: a number from the image: bone age, Cobb angle, organ volume, fetal biometry
- **outcome prediction**: a future risk, response, or survival estimate from the image (prognosis, radiomics signatures)
- **reconstruction / imputation**: a better or missing image: lower dose, faster scans, denoising, synthetic CT/MR
- **report generation / LLM**: text: draft, summarize, or extract from radiology reports
- **foundation model / vision-language**: a large pretrained model reused across tasks (segment-anything, self-supervised)
- **agent / autonomous**: multi-step actions taken without a human in the loop
- **workflow / non-interpretive**: protocoling, scheduling, ordering, decision support, education

Pediatric radiology AI publications, stratified by modality and task



Same construction, restricted to the pediatric subset (10,620 papers incl. 479 preprints). Top pediatric modalities: MRI 41%, CT 16%, ultrasound 13%, x-ray / radiography 13%. Mammography is omitted: its 79 pediatric-query hits (of 10,620) are adult breast-imaging papers whose abstracts mention children.

- **detection / diagnosis**: is a finding present now, and which one (label or box): fracture, pneumonia, tumor; includes triage and screening
- **segmentation**: outline a structure or lesion (a mask)
- **measurement / quantification**: a number from the image: bone age, Cobb angle, organ volume, fetal biometry
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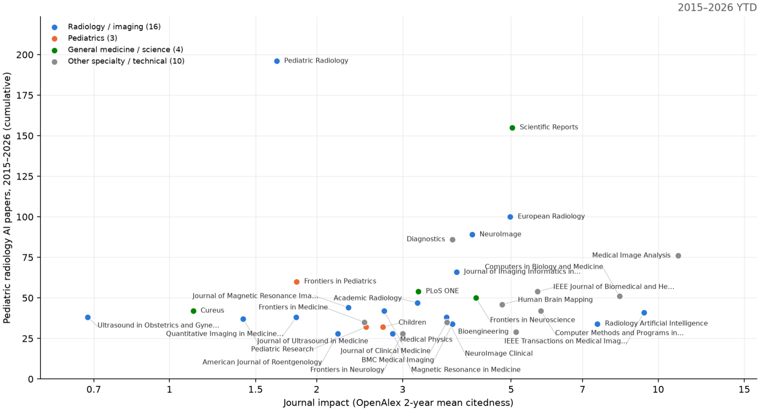
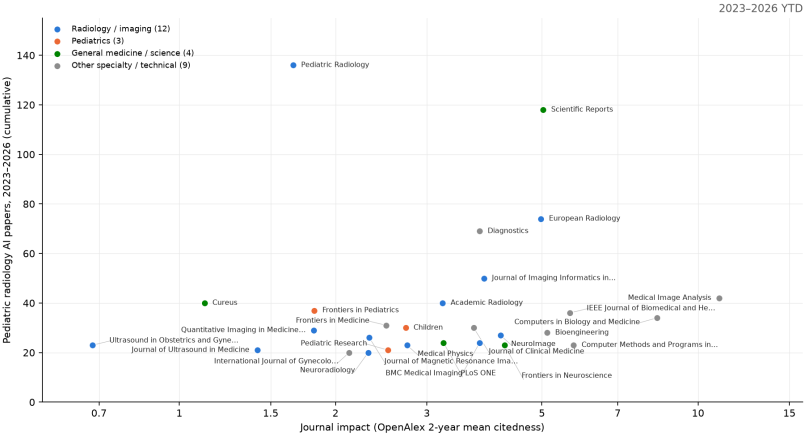
Which journals publish pediatric radiology AI



x = OpenAlex 2-year mean citedness (a free JIF analogue; runs below the published JIF). y = cumulative pediatric radiology-AI papers 2015-2026 (PubMed, strict title/abstract query). The 33 journals with at least 25 of them are shown, carrying 1,807 of 5,099; the rest is a long tail. Largest: Pediatric Radiology (196); median impact 3.7, 1 at 10 or above – this work lands mostly in mid-impact specialty and technical journals. A further 3 conference proceedings and preprint servers clear the threshold (94 papers) but have no impact number of any kind and are not plotted. The same chart animated year by year: `figures/journal_impact.gif`.

Which journals publish pediatric radiology AI, 2023–present



x = OpenAlex 2-year mean citedness (a free JIF analogue; runs below the published JIF). y = cumulative pediatric radiology-AI papers 2023-2026 (PubMed, strict title/abstract query). The 28 journals with at least 20 of them are shown, carrying 1,099 of 3,340; the rest is a long tail. Largest: Pediatric Radiology (136); median impact 3.2, 1 at 10 or above – this work lands mostly in mid-impact specialty and technical journals. medRxiv clears the threshold (31 papers) but has no impact number of any kind, so it is not plotted.

Cross-check: an independent 2025 scoping review of pediatric radiology AI

Kamran et al., *Pediatric Radiology* 2025 (doi 10.1007/s00247-025-06462-5): 789 original pediatric-focused articles hand-screened from four databases, 2005 to August 2024. Our PubMed counts are larger because they include any paper that mentions a pediatric term.

- **Modality**: radiography 38%, MRI 33%, ultrasound 14%, CT 11%, nuclear 2%. Our pediatric query ranks MRI first (MRI 41%, CT 16%, ultrasound 13%, x-ray / radiography 13%) because MRI-heavy neurodevelopment papers mention children.
- **Subspecialty**: musculoskeletal 33%, neuro 29%, chest 17%, body 14%, cardiac 6%.
- **Top applications**: bone age (88 articles), pneumonia (65), brain tumors (54), scoliosis (45), hip dysplasia (28), congenital heart disease (25), autism (18), epilepsy (14), ADHD (9), hydrocephalus (9).
- **Use of AI**: 91% image interpretation / diagnosis; 5.6% image quality; under 2% for acquisition, communication, protocoling, education, and policy combined.
- **Data**: 65% used a single local hospital dataset; 7% did not report the dataset; China 28%, USA 25%, Canada 7% of articles.
- **Growth**: 23 articles in 2018, 158 in 2022, 171 in 2023, consistent with our PubMed curve.
- **Recommendations**: multi-institutional, age-diverse pediatric datasets; CLAIM-style reporting with subgroup results and calibration; funding for the neglected non-interpretive areas (policy, education, implementation); children and families as stakeholders across the AI lifecycle.
- **What it adds to this deck**: the same three leading tasks (bone age, pneumonia, brain tumor) and the same gap (small local datasets, adult tools applied to children) reached by a manual review.

The biggest public pediatric radiology AI datasets

Dataset	Year	Modality	Size	Ages	Access
CBTN clinical MRI (Children's Brain Tumor Network)	2023	brain / spine tumor MRI	23,101 exams (1,526 patients); tumor masks for 370	pediatric	application (NCI CCDI)
GRAZPEDWRI-DX	2022	wrist radiographs (trauma)	20,327 images (6,091 patients)	0.2–19 y	open
RSNA Pediatric Bone Age Challenge	2017	hand radiographs	14,236 images (bone age + sex)	mean 10.6 y	open (RSNA terms)
FETAL_PLANES_DB	2020	fetal ultrasound (standard planes)	12,400 images (1,792 pregnancies)	fetal	open
ABCD Study	2018–	brain MRI, longitudinal	11,878 participants, up to 7 follow-ups	9–10 y at baseline	application (NBDC)
PediCXR / VinDr-PCXR	2023	chest radiographs	9,125 studies; 36 findings, 15 diagnoses	$<$ 10 y	credentialed (PhysioNet)
Guangzhou pediatric chest X-ray (Kermany)	2018	chest radiographs	5,856 images (normal / bacterial / viral pneumonia)	1–5 y	open
ABIDE I + II (autism)	2012 / 2016	brain MRI (structural, resting fMRI)	2,156 subjects	5–64 y (mostly children)	open
Regensburg Pediatric Appendicitis	2023	abdominal ultrasound + labs	1,709 images (579 patients)	0–18 y	open
dHCP (Developing Human Connectome Project)	2022–24	neonatal + fetal brain MRI	886 neonatal + 297 fetal datasets, with segmentations	20–45 weeks	open (NDA)
Pediatric-CT-SEG (TCIA)	2022	chest / abdomen / pelvis CT	718 series (359 patients); 29 organ contours	5 days – 16 y	open
BraTS-PEDs (pediatric brain tumor challenge)	2023–25	brain tumor MRI	464 patients (2024 challenge); 1,649 series on TCIA	pediatric	registration / TCIA
FeTA (fetal brain tissue annotation)	2021–24	fetal brain MRI	280 volumes (120 train + 160 test); 7 tissue labels	18–35 weeks	registration

Sizes verified against the primary paper or hosting page (2026-09). Only two pediatric radiograph sets exceed 10,000 images; the largest public pediatric-only CT set has 359 patients; no RSNA pediatric challenge has run since bone age in 2017. Adult benchmarks (CheXpert 224k, MIMIC-CXR 377k) exclude children.

Most-cited radiology AI papers, 2008-2022

Cites	FWCI	Paper	Venue	What it answers
14,326	1023.0	A survey on deep learning in medical image analysis (2017)	Medical Image Anal.	*survey*: What can deep learning do across medical imaging? (the field's reference surv...
5,658	446.6	Swin-Unet: Unet-like Pure Transformer for Medical Image Segmentation (2021)	ECCV Workshops	*segmentation*
2,990	340.4	Deep Learning in Medical Image Analysis (2017)	Annual Review of Biomedical...	*survey*: How does deep learning apply to medical image analysis? (review)
2,553	290.0	Swin UNETR: Swin Transformers for Semantic Segmentation of Brain Tumors in MR... (2022)	BrainLes@MICCAI	*segmentation*
2,769	272.4	COVID-Net: a tailored deep convolutional neural network design for detection... (2020)	Scientific Reports	*classification*: Can a CNN detect COVID-19 on chest X-ray? (COVID-Net)
3,634	236.9	ChestX-Ray8: Hospital-Scale Chest X-Ray Database and Benchmarks on Weakly-Sup... (2017)	Computer Vision and Pattern...	*classification*
10,044	226.3	nnU-Net: a self-configuring method for deep learning-based biomedical image s... (2020)	Nature Methods	*segmentation*: Can one self-configuring method segment any organ or lesion without hand tuni...
4,100	170.2	Convolutional neural networks: an overview and application in radiology (2018)	Insights into Imaging	*tutorial*: What is a convolutional neural network, explained for radiologists?
5,587	158.4	Computational Radiomics System to Decode the Radiographic Phenotype (2017)	Cancer Research	*radiomics*: Can standardized image features (radiomics) be extracted reproducibly? (PyRad...
3,322	149.0	Artificial intelligence in radiology (2018)	Nature Reviews. Cancer	*review*: How will AI change radiology practice? (Hosny et al.)

Rows are the most-cited papers of the period, ordered by FWCI = field-weighted citation impact (OpenAlex): citations relative to the world average for papers of the same field and year (1 = average), so a recent paper can be compared with an older one. Preprints (arXiv, medRxiv) are included; the venue column names the server.

Most-cited radiology AI papers, 2023

Cites	FWCI	Paper	Venue	What it answers
1,608	615.8	Segment anything in medical images	Nature Communications	*foundation*: Can a prompt-based 'segment anything' model work on medical images? (MedSAM)
2,099	243.8	Foundation models for generalist medical artificial intelligence	Nature	*foundation model*: What would a single generalist medical AI model that handles images, text and...
702	118.5	Redefining Radiology: A Review of Artificial Intelligence Integration in Medi...	Diagnostics	*survey / review*: Where does AI fit in radiology workflow, from acquisition to reporting? (revi...
509	75.2	Medical image analysis using deep learning algorithms	Frontiers in Public Health	*other*
562	74.5	A Study of CNN and Transfer Learning in Medical Imaging: Advantages, Challeng...	Sustainability	*other*
861	69.6	Segment Anything Model for medical image analysis: an experimental study	Medical Image Anal.	*segmentation*: How well does the unmodified Segment Anything Model work on medical images? (...
573	55.8	MRI-based brain tumor detection using convolutional deep learning methods and...	BMC Medical Informatics and...	*oncology*
460	44.5	Brain Tumor Detection Based on Deep Learning Approaches and Magnetic Resonanc...	Cancers	*oncology*
391	42.2	SAM-Adapter: Adapting Segment Anything in Underperformed Scenes	2023 IEEE/CVF International...	*segmentation*
442	37.4	A generalist vision–language foundation model for diverse biomedical tasks	Nature Medicine	*other*

Rows are the most-cited papers of the period, ordered by FWCI = field-weighted citation impact (OpenAlex): citations relative to the world average for papers of the same field and year (1 = average), so a recent paper can be compared with an older one. Preprints (arXiv, medRxiv) are included; the venue column names the server. 1 of the 10 rows are preprints.

Most-cited radiology AI papers, 2024

Cites	FWCI	Paper	Venue	What it answers
1,074	225.2	VM-UNet: Vision Mamba UNet for Medical Image Segmentation	ACM Transactions on Multime...	*segmentation*: Does a state-space (Mamba) backbone beat convolutions and transformers for me...
242	110.3	Generalist foundation models from a multimodal dataset for 3D computed tomogr...	Nature Biomedical Engineeri...	*other*
282	74.6	Artificial intelligence in neuro-oncology: advances and challenges in brain t...	npj Precision Oncology	*oncology*
279	72.6	METhodological RadiomICs Score (METRICS): a quality scoring tool for radiomic...	Insights into Imaging	*radiomics*: How should the quality of a radiomics study be scored? (METRICS checklist)
349	47.6	Deep learning for medical image segmentation: State-of-the-art advancements a...	Informatics in Medicine Unl...	*segmentation*
317	43.4	Advances in Medical Image Segmentation: A Comprehensive Review of Traditional...	Bioengineering	*segmentation*
238	37.0	Employing deep learning and transfer learning for accurate brain tumor detect...	Scientific Reports	*oncology*
242	36.0	Deep semi-supervised learning for medical image segmentation: A review	Expert systems with applica...	*segmentation*
241	34.5	A review of deep learning-based information fusion techniques for multimodal...	Comput. Biol. Medicine	*survey / review*
298	31.8	Vision-language models for medical report generation and visual question answ...	Frontiers Artif. Intell.	*report generation / LLM*: What can vision-language models do for report generation and visual question...

Most-cited radiology AI papers, 2025

Cites	FWCI	Paper	Venue	What it answers
258	225.8	Deep Convolutional Neural Networks in Medical Image Analysis: A Review	Inf.	*survey / review*
145	158.8	Multi-model assurance analysis showing large language models are highly vulne...	Communications Medicine	*LLM / reports*
122	108.2	Multimodal generative AI for medical image interpretation	Nature	*other*
313	94.9	Towards generalist foundation model for radiology by leveraging web-scale 2D&...	Nature Communications	*foundation model*: Can a radiology foundation model trained on web-scale 2D and 3D scans general...
173	87.2	Development and validation of an autonomous artificial intelligence agent for...	Nature Cancer	*agent / autonomous*: Can an autonomous LLM agent that calls imaging and other tools support oncolo...
208	65.9	A review of deep learning for brain tumor analysis in MRI	npj Precision Oncology	*survey / review*
150	65.8	Med-R1: Reinforcement Learning for Generalizable Medical Reasoning in Vision-...	IEEE Transactions on Medica...	*LLM / reports*
233	64.6	Artificial intelligence in healthcare and medicine: clinical applications, th...	European Journal of Medical...	*other*
151	54.0	Medical Image Segmentation: A Comprehensive Review of Deep Learning-Based Met...	Tomography	*segmentation*
118	39.5	Advanced Brain Tumor Classification in MR Images Using Transfer Learning and...	Cancers	*oncology*

1 of the 10 rows are preprints.

Most-cited radiology AI papers, 2026 (year to date)

Cites	FWCI	Paper	Venue	What it answers
48	102.6	A generalizable foundation model for analysis of human brain MRI	Nature Neuroscience	*other*
16	90.5	Reporting checklist for foundation and large language models in medical resea...	Diagnostic and Intervention...	*LLM / reports*
15	76.1	Scaling medical AI across clinical contexts	Nature Medicine	*foundation model*: How can medical AI be scaled across hospitals, populations and tasks without...
27	67.3	Applications and limitations of large language models to integrate medical co...	Iran Journal of Computer Sc...	*survey / review*
18	62.7	A hybrid deep learning framework based on VGG19 and U-Net for accurate brain...	Magnetic Resonance Letters	*segmentation*
37	58.8	Explainable artificial intelligence (XAI) in medical imaging: a systematic re...	BMC Medical Imaging	*survey / review*: Which explainability methods are used in medical imaging AI and do they help?...
12	45.3	Explainable AI-driven MRI-based brain tumor classification: a novel deep lear...	Frontiers Artif. Intell.	*oncology*
21	44.2	Agentic AI in Radiology: Evolution from Large Language Models to Future Clini...	Radiology: Artificial Intel...	*agent / autonomous*: What are agentic AI systems in radiology and what stands between them and cli...
17	28.4	Leveraging multi-modal foundation models for analysing spatial multi-omic and...	Nature Biomedical Engineeri...	*other*
19	21.4	BIBSNet: A deep learning baby image brain segmentation network for MRI scans	Developmental Cognitive Neu...	*segmentation*: Can one network segment infant brain MRI across the first years of life? (BIB...

Citations for the current year are still accumulating; the FWCI column is the more stable signal here. 1 of the 10 rows are preprints.

Most-cited pediatric radiology AI papers, 2008-2022

Cites	FWCI	Paper	Venue	What it answers
426	328.6	Performance of a Deep-Learning Neural Network Model in Assessing Skeletal Mat... (2017)	Radiology	*bone age*: Can a CNN assess skeletal maturity as accurately as radiologists?
431	321.5	Fully Automated Deep Learning System for Bone Age Assessment (2017)	Journal of digital imaging	*bone age*: Fully automated bone age from hand radiographs
403	221.2	Deep learning for automated skeletal bone age assessment in X-ray images (2017)	Medical Image Anal.	*bone age*: Deep learning for automated bone age (BoNet)
143	73.0	Computerized Bone Age Estimation Using Deep Learning Based Program: Evaluatio... (2017)	AJR. American journal of ro...	*bone age*: Is a commercial deep-learning bone-age program accurate and faster in practic...
191	57.8	A Review on Deep-Learning Algorithms for Fetal Ultrasound-Image Analysis (2022)	Medical Image Anal.	*survey / review*
352	57.4	Standard Plane Localization in Fetal Ultrasound via Domain Transferred Deep N... (2015)	IEEE journal of biomedical...	*fetal MRI*
194	57.0	Automatic Fetal Ultrasound Standard Plane Recognition Based on Deep Learning... (2021)	IEEE Transactions on Indust...	*fetal MRI*
233	30.3	Evaluation of deep convolutional neural networks for automatic classification... (2020)	Scientific Reports	*fetal MRI*
458	27.0	The RSNA Pediatric Bone Age Machine Learning Challenge. (2019)	Radiology	*bone age*: How well can crowdsourced AI estimate bone age? (RSNA challenge, 260 teams)
253	25.8	An ensemble of neural networks provides expert-level prenatal detection of co... (2021)	Nature Medicine	*detection*

Most-cited pediatric radiology AI papers, 2023

Cites	FWCI	Paper	Venue	What it answers
32	36.6	Bone Age Assessment Using Artificial Intelligence in Korean Pediatric Populat...	Korean Journal of Radiology	*bone age*
62	33.6	Standard fetal ultrasound plane classification based on stacked ensemble of d...	Expert systems with applica...	*fetal ultrasound*: Can an ensemble of networks classify the standard fetal ultrasound planes?
27	33.5	Bone age assessment based on deep neural networks with annotation-free cascad...	Frontiers in Artificial Int...	*bone age*
43	23.8	Review on deep learning fetal brain segmentation from Magnetic Resonance imag...	Artif. Intell. Medicine	*segmentation*
37	20.2	Transfer learning for accurate fetal organ classification from ultrasound ima...	Scientific Reports	*fetal MRI*
34	16.6	Deep Learning-Based Multiclass Brain Tissue Segmentation in Fetal MRIs	Italian National Conference...	*segmentation*
43	9.7	Deeplasia: deep learning for bone age assessment validated on skeletal dyspla...	Pediatric Radiology	*bone age*
30	7.8	Deep learning for estimation of fetal weight throughout the pregnancy from fe...	American Journal of Obstetr...	*fetal MRI*
38	7.3	Deep learning model for prenatal congenital heart disease (CHD) screening can...	Ultrasound in Obstetrics an...	*fetal ultrasound*: Does a prenatal congenital heart disease screening model work on retrospectiv...
31	5.8	MRI-Based End-To-End Pediatric Low-Grade Glioma Segmentation and Classificati...	Canadian Association of Rad...	*segmentation*

Rows are the most-cited papers of the period, ordered by FWCI = field-weighted citation impact (OpenAlex): citations relative to the world average for papers of the same field and year (1 = average), so a recent paper can be compared with an older one. Preprints (arXiv, medRxiv) are included; the venue column names the server. 1 of the 10 rows are preprints.

Most-cited pediatric radiology AI papers, 2024

Cites	FWCI	Paper	Venue	What it answers
17	34.8	Intrapartum Ultrasound Image Segmentation of Pubic Symphysis and Fetal Head U...	International Conference on...	*segmentation*
30	23.8	FetalBrainAwareNet: Bridging GANs with anatomical insight for fetal ultrasoun...	Comput. Medical Imaging Gra...	*fetal MRI*
20	19.5	Artificial intelligence assisted common maternal fetal planes prediction from...	Frontiers in Medicine	*fetal MRI*
18	15.9	Deep learning denoising reconstruction for improved image quality in fetal ca...	Frontiers in Cardiovascular...	*reconstruction*: Does deep-learning denoising improve fetal cardiac cine MRI?
30	15.1	RTSeg-net: A lightweight network for real-time segmentation of fetal head and...	Comput. Biol. Medicine	*segmentation*
16	13.4	FetSAM: Advanced Segmentation Techniques for Fetal Head Biometrics in Ultraso...	IEEE Open Journal of Engine...	*segmentation*
23	7.0	Deep learning-based automated lesion segmentation on pediatric focal cortical...	Insights into Imaging	*epilepsy*: Can focal cortical dysplasia lesions be found automatically on preoperative p...
18	5.3	Integration of a Deep Convolutional Neural Network With Adaptive Channel Weig...	IEEE Transactions on Instru...	*fetal MRI*
18	5.2	Explainable AI in Deep Learning Based Classification of Fetal Ultrasound Imag...	Procedia Computer Science	*fetal MRI*
19	4.8	A deep learning framework for identifying and segmenting three vessels in fet...	BioMedical Engineering OnLi...	*segmentation*

Most-cited pediatric radiology AI papers, 2025

Cites	FWCI	Paper	Venue	What it answers
16	171.3	A comprehensive review of artificial intelligence - based algorithm towards f...	Artificial Intelligence Rev...	*survey / review*
18	85.1	M-CNN-RF: A hybrid deep learning model for accurate pediatric skeletal age es...	Alexandria Engineering Jour...	*other*
20	77.0	Bone Age Assessment Using Various Medical Imaging Techniques Enhanced by Arti...	Diagnostics	*bone age*
18	24.9	Advancing prenatal healthcare by explainable AI enhanced fetal ultrasound ima...	Scientific Reports	*segmentation*
12	21.8	FetalMovNet: A Novel Deep Learning Model Based on Attention Mechanism for Fet...	IEEE Access	*fetal MRI*
11	14.5	Effectiveness and clinical impact of using deep learning for first-trimester...	BMC Pregnancy and Childbirth	*fetal MRI*
12	12.4	Association between deep learning radiomics based on placental MRI and preecl...	European Journal of Radiolo...	*radiomics*
16	8.7	Prenatal detection of congenital heart defects using the deep learning-based...	BMJ Open	*fetal ultrasound*: Can deep learning on prenatal ultrasound video detect congenital heart defect...
11	8.6	Multi-view deep learning framework for the detection of chest X-rays compatib...	Nature Communications	*detection*
11	7.0	Evaluation of Image Quality and Scan Time Efficiency in Accelerated 3D T1-Wei...	Korean Journal of Radiology	*reconstruction*

Most-cited pediatric radiology AI papers, 2026 (year to date)

Cites	FWCI	Paper	Venue	What it answers
11	34.3	Artificial intelligence and radiomics for pediatric brain tumor classificatio...	Neuroradiology	*oncology*: Can AI and radiomics classify pediatric brain tumors and their molecular subt...
5	27.9	Deep learning assessment of fetal brain maturation on 3D ultrasound volumes i...	Ultrasound in Obstetrics an...	*fetal ultrasound*: Can deep learning grade fetal brain maturation on 3D ultrasound in growth res...
3	17.6	Foundation models in radiology: a primer for pediatric radiologists	Pediatric Radiology	*foundation model*: What should a pediatric radiologist know about foundation models? (primer)
2	17.6	Deep Learning-Based Multi-Class Pediatric Wrist Fracture Subtype Classificati...	Journal of Imaging	*fracture*
3	17.3	Current applications and challenges of artificial intelligence applied to dia...	Pediatric Radiology	*classification*
4	13.6	Data mining in pediatric radiology in the era of artificial intelligence.	Pediatric Radiology	*other*
2	13.2	Acquisition time/dose reduction in pediatric PET imaging using patch-based de...	EJNMMI Physics	*reconstruction*: Can pediatric PET use one fifth of the counts with patch-based deep-learning...
2	12.6	Optimizing pediatric chest CT: superior image quality and lower radiation dos...	European Journal of Radiolo...	*reconstruction*: Can deep-learning reconstruction lower pediatric chest CT dose while improvin...
2	12.2	Enhancing efficiency in pediatric brain tumor segmentation using a pathologic...	Neuro-Oncology Advances	*segmentation*
3	11.7	Automated bone age assessment in rare pediatric growth disorders: a comparati...	Frontiers in Endocrinology	*bone age*: Does an open bone-age model hold up in rare growth disorders against several...

Citations for the current year are still accumulating; the FWCI column is the more stable signal here.

NeurIPS: radiology AI works, 2023–2026

Conference on Neural Information Processing Systems. Radiology-AI works found, 2023–2026: 45 (2023: 13, 2024: 12, 2025: 20); 0 pediatric (marked ⋆).

Year	Cites	FWCI	Paper
2025	24	–	3D-RAD: A Comprehensive 3D Radiology Med-VQA Dataset with Multi-Temporal Analysis and Diverse Diagnostic Tasks
2025	20	–	Better Tokens for Better 3D: Advancing Vision-Language Modeling in 3D Medical Imaging
2024	47	2.0	Touchstone Benchmark: Are We on the Right Way for Evaluating AI Algorithms for Medical Segmentation?
2024	26	0.9	Eye-gaze Guided Multi-modal Alignment for Medical Representation Learning
2024	25	1.2	Enhancing vision-language models for medical imaging: bridging the 3D gap with innovative slice selection
2024	20	–	DiffusionBlend: Learning 3D Image Prior through Position-aware Diffusion Score Blending for 3D Computed Tomography Reco...
2023	144	–	SegVol: Universal and Interactive Volumetric Medical Image Segmentation
2023	104	–	LVM-Med: Learning Large-Scale Self-Supervised Vision Models for Medical Imaging via Second-order Graph Matching
2023	72	–	EHRXQA: A Multi-Modal Question Answering Dataset for Electronic Health Records with Chest X-ray Images
2023	66	–	SwiFT: Swin 4D fMRI Transformer
2023	64	–	NeuroGraph: Benchmarks for Graph Machine Learning in Brain Connectomics
2023	19	–	Knowledge-based in silico models and dataset for the comparative evaluation of mammography AI for a range of breast cha...

Source: Semantic Scholar venue search, citations and FWCI from OpenAlex; title-level relevance filter; the 12 most-cited of 45 works, newest year first.

ICLR: radiology AI works, 2023–2026

International Conference on Learning Representations. Radiology-AI works found, 2023–2026: 17 (2023: 4, 2024: 5, 2025: 8); 0 pediatric (marked ⋆).

Year	Cites	FWCI	Paper
2025	70	–	Large-scale and Fine-grained Vision-language Pre-training for Enhanced CT Image Understanding
2025	13	–	SimXRD-4M: Big Simulated X-ray Diffraction Data and Crystal Symmetry Classification Benchmark
2025	9	–	Self-Supervised Diffusion MRI Denoising via Iterative and Stable Refinement
2025	9	–	Score-based Self-supervised MRI Denoising
2024	143	–	MMed-RAG: Versatile Multimodal RAG System for Medical Vision Language Models
2024	31	–	MediConfusion: Can you trust your AI radiologist? Probing the reliability of multimodal medical foundation models
2024	19	–	The Effect of Intrinsic Dataset Properties on Generalization: Unraveling Learning Differences Between Natural and Medic...
2024	16	–	LeFusion: Controllable Pathology Synthesis via Lesion-Focused Diffusion Models
2023	107	–	Advancing Radiograph Representation Learning with Masked Record Modeling
2023	84	–	DDM2: Self-Supervised Diffusion MRI Denoising with Generative Diffusion Models
2023	40	–	AE-FLOW: Autoencoders with Normalizing Flows for Medical Images Anomaly Detection
2023	18	–	Improved Training of Physics-Informed Neural Networks Using Energy-Based Priors: a Study on Electrical Impedance Tomogr...

Source: Semantic Scholar venue search, citations and FWCI from OpenAlex; title-level relevance filter; the 12 most-cited of 17 works, newest year first.

ICML: radiology AI works, 2023–2026

International Conference on Machine Learning. Radiology-AI works found, 2023–2026: 12 (2023: 4, 2024: 2, 2025: 6); 0 pediatric (marked $\star$).

Year	Cites	FWCI	Paper
2025	21	–	CLIMB: Data Foundations for Large Scale Multimodal Clinical Foundation Models
2025	17	–	MindLLM: A Subject-Agnostic and Versatile Model for fMRI-to-Text Decoding
2025	11	–	Raptor: Scalable Train-Free Embeddings for 3D Medical Volumes Leveraging Pretrained 2D Foundation Models
2025	5	–	Efficient Noise Calculation in Deep Learning-based MRI Reconstructions
2025	2	–	LangDAug: Langevin Data Augmentation for Multi-Source Domain Generalization in Medical Image Segmentation
2025	0	–	Staged and Physics-Grounded Learning Framework with Hyperintensity Prior for Pre-Contrast MRI Synthesis
2024	14	–	Unsupervised Domain Adaptation for Anatomical Structure Detection in Ultrasound Images
2024	10	–	Adaptive Sampling of k-Space in Magnetic Resonance for Rapid Pathology Prediction
2023	17	–	ACAT: Adversarial Counterfactual Attention for Classification and Detection in Medical Imaging
2023	16	–	Robustness of Deep Learning for Accelerated MRI: Benefits of Diverse Training Data
2023	15	–	Learning High-Order Relationships of Brain Regions
2023	8	–	Contextualized Policy Recovery: Modeling and Interpreting Medical Decisions with Adaptive Imitation Learning

Source: Semantic Scholar venue search, citations and FWCI from OpenAlex; title-level relevance filter; the 12 most-cited of 12 works, newest year first.

CVPR: radiology AI works, 2023–2026

IEEE/CVF Conference on Computer Vision and Pattern Recognition. Radiology-AI works found, 2023–2026: 57 (2023: 15, 2024: 24, 2025: 18); 1 pediatric (marked $\star$).

Year	Cites	FWCI	Paper
2025	68	265.4	Enhanced Contrastive Learning with Multi-view Longitudinal Data for Chest X-ray Report Generation
2024	84	47.9	Towards Generalizable Tumor Synthesis
2024	81	14.6	Rethinking Diffusion Model for Multi-Contrast MRI Super-Resolution
2024	73	13.8	MemSAM: Taming Segment Anything Model for Echocardiography Video Segmentation
2023	243	20.2	Dynamic Graph Enhanced Contrastive Learning for Chest X-Ray Report Generation
2023	222	75.5	Ambiguous Medical Image Segmentation Using Diffusion Models
2023	217	18.7	METransformer: Radiology Report Generation by Transformer with Multiple Learnable Expert Tokens
2023	164	21.9	MCF: Mutual Correction Framework for Semi-Supervised Medical Image Segmentation
2023	133	14.4	KiUT: Knowledge-injected U-Transformer for Radiology Report Generation
2023	117	67.2	Structure-Aware Sparse-View X-Ray 3D Reconstruction
2023	103	11.0	Label-Free Liver Tumor Segmentation
2023	75	14.1	Learning Federated Visual Prompt in Null Space for MRI Reconstruction

Source: Semantic Scholar venue search, citations and FWCI from OpenAlex; title-level relevance filter; the 12 most-cited of 57 works, newest year first.

MICCAI: radiology AI works, 2023–2026

Medical Image Computing and Computer-Assisted Intervention. Radiology-AI works found, 2023–2026: 521 (2023: 139, 2024: 195, 2025: 187); 25 pediatric (marked ⋆).

Year	Cites	FWCI	Paper
2025	208	1.1	MedVLM-R1: Incentivizing Medical Reasoning Capability of Vision-Language Models (VLMs) via Reinforcement Learning
2024	133	–	CT2Rep: Automated Radiology Report Generation for 3D Medical Imaging
2024	102	–	Anatomically-Controllable Medical Image Generation with Segmentation-Guided Diffusion Models
2024	87	–	MedCLIP-SAM: Bridging Text and Image Towards Universal Medical Image Segmentation
2023	462	–	MedNeXt: Transformer-driven Scaling of ConvNets for Medical Image Segmentation
2023	128	–	CoLa-Diff: Conditional Latent Diffusion Model for Multi-Modal MRI Synthesis
2023	105	–	DiffMIC: Dual-Guidance Diffusion Network for Medical Image Classification
2023	97	13.9	Beyond Adapting SAM: Towards End-to-End Ultrasound Image Segmentation via Auto Prompting
2023	95	–	Self-Supervised MRI Reconstruction with Unrolled Diffusion Models
2023	83	–	Make-A-Volume: Leveraging Latent Diffusion Models for Cross-Modality 3D Brain MRI Synthesis
2023	72	19.0	BerDiff: Conditional Bernoulli Diffusion Model for Medical Image Segmentation
2023	71	–	Ariadne's Thread: Using Text Prompts to Improve Segmentation of Infected Areas from Chest X-ray images

Source: Semantic Scholar venue search, citations and FWCI from OpenAlex; title-level relevance filter; the 12 most-cited of 521 works, newest year first.

MIDL: radiology AI works, 2023–2026

Medical Imaging with Deep Learning. Radiology-AI works found, 2023–2026: 62 (2023: 33, 2024: 29); 2 pediatric (marked ⋆).

Year	Cites	FWCI	Paper
2024	14	–	Imbalance-aware loss functions improve medical image classification
2023	78	–	Patched Diffusion Models for Unsupervised Anomaly Detection in Brain MRI
2023	39	–	MMCFormer: Missing Modality Compensation Transformer for Brain Tumor Segmentation
2023	29	–	Medical diffusion on a budget: textual inversion for medical image generation
2023	23	–	MProtoNet: A Case-Based Interpretable Model for Brain Tumor Classification with 3D Multi-parametric Magnetic Resonance...
2023	22	–	Diffusion Models for Contrast Harmonization of Magnetic Resonance Images
2023	20	–	Exploring Image Augmentations for Siamese Representation Learning with Chest X-Rays
2023	19	–	3D Medical Axial Transformer: A Lightweight Transformer Model for 3D Brain Tumor Segmentation
2023	19	–	Vision-Language Modelling For Radiological Imaging and Reports In The Low Data Regime
2023	14	–	E(3) x SO(3)-Equivariant Networks for Spherical Deconvolution in Diffusion MRI
2023	14	–	Amortized Normalizing Flows for Transcranial Ultrasound with Uncertainty Quantification
2023	13	–	Selective experience replay compression using coresets for lifelong deep reinforcement learning in medical imaging

Source: Semantic Scholar venue search, citations and FWCI from OpenAlex; title-level relevance filter; the 12 most-cited of 62 works, newest year first.

RSNA: radiology AI works, 2023–2026

Radiological Society of North America: Radiology, Radiology: AI, RadioGraphics. Radiology-AI works found, 2023–2026: 616 (2023: 151, 2024: 174, 2025: 185, 2026: 106); 13 pediatric (marked ⋆). RSNA annual-meeting abstracts are not indexed; the society's journals stand in.

Year	Cites	FWCI	Paper
2024	452	113.7	Checklist for Artificial Intelligence in Medical Imaging (CLAIM): 2024 Update.
2024	324	26.4	Chatbots and Large Language Models in Radiology: A Practical Primer for Clinical and Research Applications.
2024	215	58.7	The Image Biomarker Standardization Initiative: Standardized Convolutional Filters for Reproducible Radiomics and Enhan...
2023	919	29.4	ChatGPT and Other Large Language Models Are Double-edged Swords.
2023	419	13.5	Performance of ChatGPT on a Radiology Board-style Examination: Insights into Current Strengths and Limitations.
2023	392	33.2	Deep Learning Image Reconstruction for CT: Technical Principles and Clinical Prospects.
2023	282	9.1	Automation Bias in Mammography: The Impact of Artificial Intelligence BI-RADS Suggestions on Reader Performance.
2023	273	44.4	Potential of ChatGPT and GPT-4 for Data Mining of Free-Text CT Reports on Lung Cancer.
2023	265	37.0	MRI-based Quantification of Intratumoral Heterogeneity for Predicting Treatment Response to Neoadjuvant Chemotherapy in...
2023	259	47.2	How AI Responds to Common Lung Cancer Questions: ChatGPT vs Google Bard.
2023	258	35.5	A Guide to Cross-Validation for Artificial Intelligence in Medical Imaging.
2023	225	36.2	Predicting Microvascular Invasion in Hepatocellular Carcinoma Using CT-based Radiomics Model.

Source: Semantic Scholar venue search, citations and FWCI from OpenAlex; title-level relevance filter; the 12 most-cited of 616 works, newest year first.

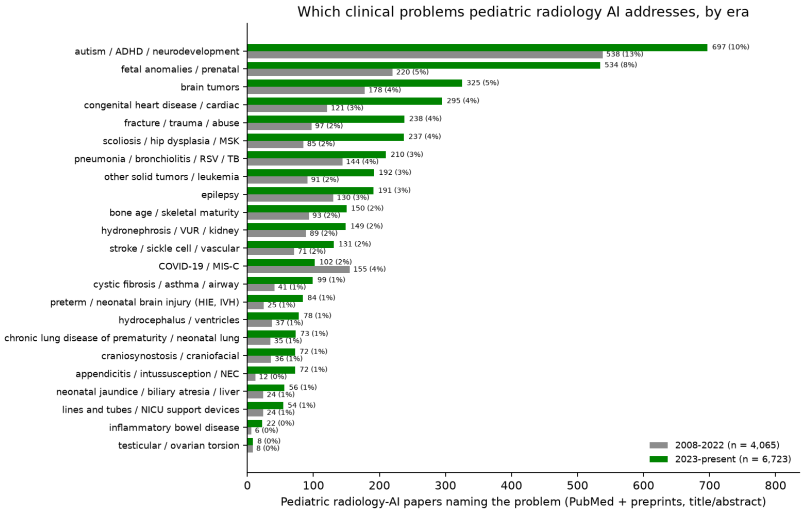
SPR: radiology AI works, 2023–2026

Society for Pediatric Radiology: Pediatric Radiology (journal). Radiology-AI works found, 2023–2026: 120 (2023: 27, 2024: 17, 2025: 37, 2026: 39); 70 pediatric (marked ⋆). SPR meeting abstracts are not indexed; the society's journal stands in.

Year	Cites	FWCI	Paper
2025	18	4.6	AI implementation in pediatric radiology for patient safety: a multi-society statement from the ACR, ESPR, SPR, SLARP,... ⋆
2025	17	27.9	Scanner-based real-time three-dimensional brain + body slice-to-volume reconstruction for T2-weighted 0.55-T low-field... ⋆
2024	24	2.0	Capability of multimodal large language models to interpret pediatric radiological images ⋆
2024	15	–	Artificial Intelligence vs. Doctors: Diagnosing Necrotizing Enterocolitis on Abdominal Radiographs
2023	49	1.6	Detecting pediatric wrist fractures using deep-learning-based object detection ⋆
2023	44	1.2	Artificial intelligence to identify fractures on pediatric and young adult upper extremity radiographs ⋆
2023	43	9.7	Deeplasia: deep learning for bone age assessment validated on skeletal dysplasias ⋆
2023	42	1.2	Comparison of diagnostic performance of a deep learning algorithm, emergency physicians, junior radiologists and senior...
2023	35	1.0	Artificial intelligence-based detection of paediatric appendicular skeletal fractures: performance and limitations for... ⋆
2023	31	1.0	The unintended consequences of artificial intelligence in paediatric radiology ⋆
2023	23	4.0	Artificial Intelligence in Paediatric Tuberculosis ⋆
2023	20	4.2	Innovative advances in pediatric radiology: computed tomography reconstruction techniques, photon-counting detector com... ⋆

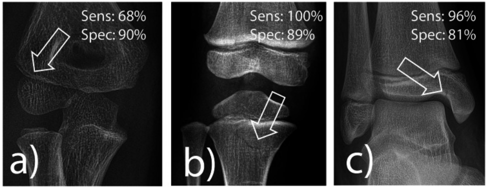
Source: Semantic Scholar venue search, citations and FWCI from OpenAlex; title-level relevance filter; the 12 most-cited of 120 works, newest year first.

Which medical problems pediatric radiology AI addresses



AI fracture detection in a real pediatric emergency department

Ziegner et al., European Radiology 2025 (second most-cited pediatric paper of 2025)



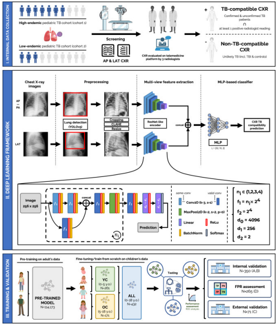
Ziegner et al., Eur Radiol 2025, figure 4: pediatric fractures the software found, with its per-region sensitivity and specificity

source: cdn.ncbi.nlm.nih.gov

- **Problem**: fractures in children are missed most often by the least experienced readers, at night, in the emergency department.
- **Data**: 1,672 consecutive radiographs of children under 18 from one tertiary pediatric emergency department, plus a selected medicolegal set (proximal tibia, radial condyle); three pediatric residents read each before and after seeing the AI.
- **Method**: a commercially available, CE-marked deep-learning fracture detector (adult-derived, pediatric extension) run stand-alone and as a second reader.
- **Result**: stand-alone sensitivity 92%, specificity 83%; 100% for proximal tibia but 68% for radial condyle fractures; residents' accuracy rose from 88% to 90%, and in 2% of cases they dropped a correct diagnosis after seeing the AI.
- **Why it matters**: the first real-life pediatric ED evaluation: the gain from a cleared adult tool is real but modest, and the elbow remains the weak spot.

pTBLightNet: pediatric tuberculosis on frontal and lateral chest radiographs

Capellán-Martín et al., Nature Communications 2025 (open access)



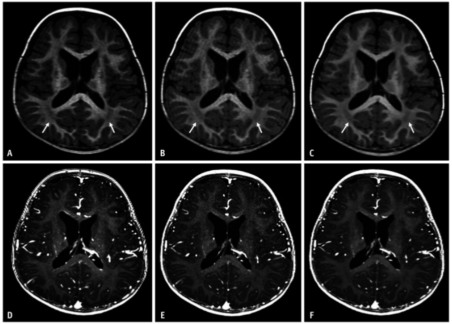
Capellan-Martin et al., Nat Commun 2025, figure 1: the multi-view pTBLightNet pipeline

source: cdn.ncbi.nlm.nih.gov

- **Problem**: pediatric pulmonary TB has vaguer symptoms and subtler radiographs than adult TB; most affected children in high-burden settings are never diagnosed.
- **Data**: pre-trained on 114,173 adult chest radiographs, then fine-tuned and tested on 918 radiographs from three pediatric TB cohorts, with lateral views.
- **Method**: a multi-view network that scores frontal and lateral films for TB-compatible findings; separate models for under-5 and 5-18 years.
- **Result**: AUC 0.90 on internal and 0.68 on external testing; lateral views helped most in the youngest children; age-specific models matched larger undifferentiated ones.
- **Why it matters**: shows both the value of adult pre-training for a rare pediatric target and the internal-to-external drop that every pediatric deployment should expect.

Deep-learning reconstruction shortens pediatric brain MRI

Yoo et al., Korean Journal of Radiology 2025 (open access)



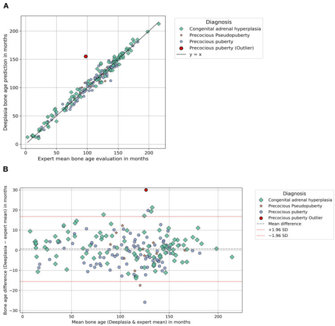
Yoo et al., Korean J Radiol 2025, figure 2: conventional (A, D) vs accelerated 3D T1 pediatric brain MRI without (B, E) and with (C, F) deep-learning reconstruction

source: cdn.ncbi.nlm.nih.gov

- **Problem**: long 3D T1 sequences are where children move, sedation is prescribed, and scanner time is lost.
- **Data**: 46 children scanned with both the conventional and an accelerated 3D T1 protocol (pre- and post-contrast) on one 3T scanner.
- **Method**: vendor deep-learning reconstruction applied to the accelerated acquisition; noise, contrast, signal-to-noise and radiologist ratings compared with the conventional scan.
- **Result**: acquisition time cut by 29% (pre-contrast) and 41% (post-contrast); the deep-learning images had lower noise, higher SNR and CNR, better overall quality and fewer artifacts than the conventional scan.
- **Why it matters**: reconstruction AI is on the scanner today and gives children a shorter, better exam without any diagnostic model in the loop.

Deeplasia bone age in rare growth disorders, against multiple experts

Skaf et al., Frontiers in Endocrinology 2026 (open access); external validation of Rassmann et al., Pediatr Radiol 2023



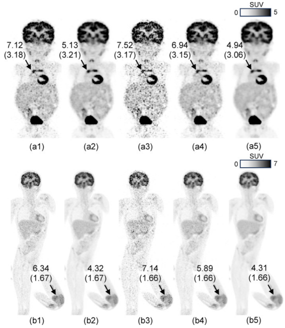
Skaf et al., Front Endocrinol 2026, figure 1: Deeplasia bone age vs the expert mean in the endocrine cohort (scatter and Bland-Altman)

source: cdn.ncbi.nlm.nih.gov

- **Problem**: bone-age models are trained on ordinary hands; the children who need bone age most have syndromes, endocrine disease or lysosomal storage disorders that distort the skeleton.
- **Data**: 1,138 hand radiographs from several centers: SHOX deficiency, Noonan, Silver-Russell, Turner, CAH, precocious puberty (cohort 1) and mucopolysaccharidoses and other storage disorders (cohort 2); two to five expert Greulich-Pyle readings each.
- **Method**: the open-source Deeplasia system run unchanged; each human rater and the model scored against the remaining experts.
- **Result**: mean absolute error 6.0 months (cohort 1) and 7.1 months (storage disorders), 1-year accuracy 90% and 81%; the model beat every individual human rater.
- **Why it matters**: an open pediatric model validated on the hardest cases, by an outside group, on external data: the template for how a children's hospital should test any tool.

Pediatric PET at one fifth of the counts with patch-based deep learning

Han et al., EJNMMI Physics 2026 (open access; Cincinnati Children's)



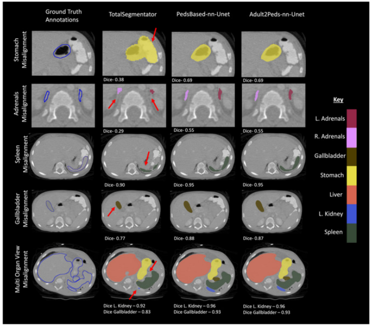
Han et al., EJNMMI Phys 2026, figure 5: pediatric whole-body FDG PET, standard 90 s vs 20 s per bed, with and without patch-based deep-learning denoising

source: cdn.ncbi.nlm.nih.gov

- **Problem**: PET dose and scan time in children are set by noise; conventional denoising networks need large training sets that pediatric PET does not have.
- **Data**: one high-quality pediatric exam for training; a NEMA phantom and 88 clinical whole-body FDG exams (under 12 years) with list-mode data truncated from 90 s to 20 s per bed.
- **Method**: a patch-based network that learns local structure from a single exam and denoises the reduced-count images; blinded observer study and SUV analysis.
- **Result**: 20-second images rated equal or better than the standard 90-second images in more than 85% of cases, with resolution, contrast and SUV preserved.
- **Why it matters**: either a 4.5x shorter scan or about 22% of today's injected activity, from a method trainable on a single local exam.

Children are not small adults: an adult CT segmentation model in children

Chatterjee et al., Journal of Imaging Informatics in Medicine 2025



Chatterjee et al., J Imaging Inform Med 2025, figure 7:

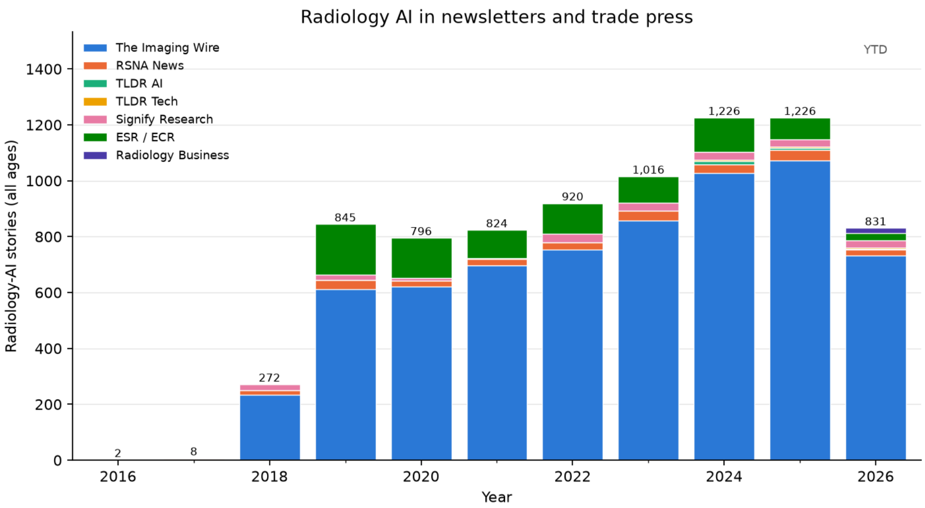
pediatric CT organ labels, ground truth vs adult-trained

TotalSegmentator vs pediatric-trained models

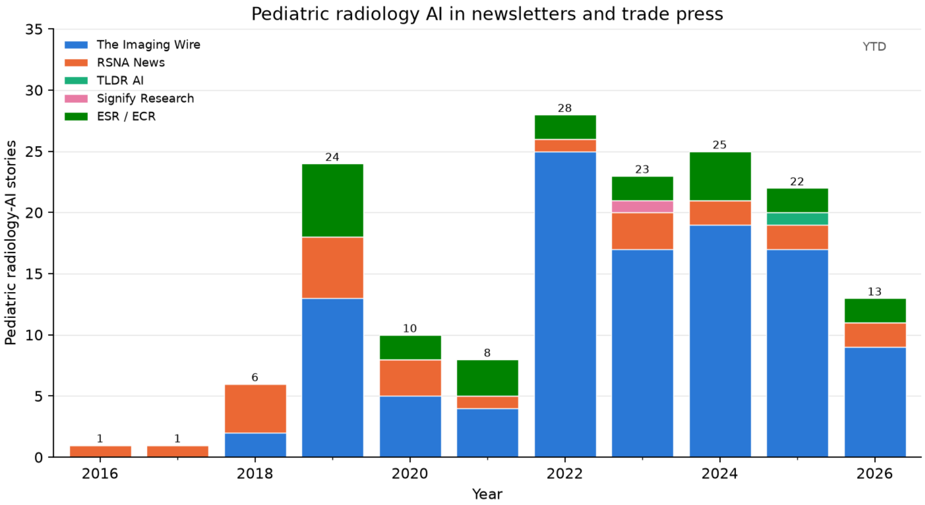
source: cdn.ncbi.nlm.nih.gov

- **Problem**: the most-used open CT organ segmentation model (TotalSegmentator) was trained on adults; nobody had measured how it behaves in children.
- **Data**: external adult (n = 300) and pediatric (n = 359) abdominal CT sets with expert organ labels.
- **Method**: TotalSegmentator scored on both sets; then a pediatric-only nnU-Net and an adult model fine-tuned on pediatric cases.
- **Result**: mean Dice fell from 0.81 in adults to 0.73 in children, worst for the adrenals (0.69 to 0.41 and 0.35), duodenum and pancreas; pediatric training or fine-tuning recovered the loss.
- **Why it matters**: the clearest quantitative case for local pediatric validation and fine-tuning of adult-derived tools before clinical use.

Newsletters — radiology AI stories, all ages



Newsletters — pediatric radiology AI stories



Pediatric radiology AI in the trade press, 2023 (23 stories)

Date	Newsletter	Story	Paper
2023-01-08	The Imaging Wire	Gestational Age AI	
2023-03-01	The Imaging Wire	Fetal Ultrasound AI Generalizability	Sendra-Balcells et al., 2023 (Scientific Re...
2023-03-08	ESR / ECR	ECR 2023 – A true celebration of the cycle of life.	
2023-03-17	The Imaging Wire	When Sao Paolo's Diagnosticos da America (DASA, the world's fourth larg...	
2023-04-07	The Imaging Wire	Ultrasound Spots Breech Pregnancies	Knights et al., 2023 (PLOS Medicine)
2023-04-07	The Imaging Wire	When Sao Paolo's Diagnosticos da America SA (DASA, the world's fourth l...	
2023-04-12	Signify Research	Where do opportunities exist?	
2023-04-12	RSNA News	RSNA Webinar to Examine Impact of Large Language Models	
2023-05-07	The Imaging Wire	New C-Arm Launched	
2023-06-07	The Imaging Wire	BrightHeart's Fetal Heart AI	
2023-06-12	RSNA News	Understanding the Imaging Spectrum of non-Hodgkin Lymphoma in Children,...	
2023-06-19	The Imaging Wire	Better Together at SIIM	
2023-06-25	The Imaging Wire	AI Detects Child Abuse on CT	
2023-07-07	RSNA News	Survivors of Childhood Cancer: Risk of Future Disease, Screening Needs...	
2023-08-02	The Imaging Wire	MRI Radiomics Hones in on Crohn's	Liu et al., 2024 (American Journal of Roent...
2023-08-02	The Imaging Wire	Sonio Gets FDA Nod for Fetal AI	
2023-09-20	The Imaging Wire	GE Lands $44M AI Ultrasound Grant	
2023-10-08	The Imaging Wire	ChatGPT Reports for Patients	Jeblick et al., 2023 (European Radiology)
2023-11-08	The Imaging Wire	Philips to Expand AI-Powered POCUS	
2023-11-08	The Imaging Wire	Stories of Success in Cloud and AI	
2023-11-12	The Imaging Wire	Uneven Success Against Breast Cancer | Pediatric Radiation	
2023-11-26	The Imaging Wire	AI of MRI Diagnoses Autism	
2023-12-11	ESR / ECR	Revolutionising paediatric radiology: the future impact of AI	

Pediatric radiology AI in the trade press, 2024 (25 stories)

Date	Newsletter	Story	Paper
2024-01-10	The Imaging Wire	AI Models Go Head-to-Head in Project AIR Study	van Leeuwen et al., 2024 (Radiology)
2024-01-15	ESR / ECR	Hidden health risks of endocrine disruptors?	
2024-01-17	The Imaging Wire	Echo AI's RHD Impact	Brown et al., 2024 (Journal of the American...
2024-01-20	The Imaging Wire	Us2.ai	
2024-02-07	The Imaging Wire	Meet Gleamer at ECR 2024	
2024-03-20	The Imaging Wire	AI Helps Residents Detect Fractures	
2024-04-22	The Imaging Wire	GE Launches US Scanners	
2024-05-19	The Imaging Wire	Early Preeclampsia Risk Screening	Guerby et al., 2024 (Hypertension)
2024-06-29	The Imaging Wire	FDA Clears New Clarius Prenatal App	
2024-07-01	ESR / ECR	AI applications in musculoskeletal imaging	
2024-07-10	RSNA News	Experts Outline Considerations to Deploy AI in Radiology	
2024-08-14	The Imaging Wire	Cardiac Ultrasound Scanner Cleared	
2024-08-25	The Imaging Wire	When Follow-Up Falls Short | Pediatric CT Use	
2024-08-25	The Imaging Wire	MAUI Emerges from Stealth	
2024-09-04	The Imaging Wire	Samsung Finalizes Sonio Acquisition	
2024-09-08	The Imaging Wire	AZmed AI Extended to Kids	
2024-10-03	ESR / ECR	On Artificial Intelligence: An interview with Susan Shelmerdine	
2024-10-06	The Imaging Wire	Fracture AI Improves Over Time	
2024-10-10	The Imaging Wire	Low-Dose CT in Kids | Hinton Nabs Nobel, Cancels MRI	
2024-10-10	The Imaging Wire	Low-Dose CT Confounds CAD in Kids	Hardie et al., 2025 (American Journal of Ro...
2024-10-17	ESR / ECR	On Artificial Intelligence: An interview with Brendan Kelly	
2024-11-19	RSNA News	Incorrect AI Advice Influences Diagnostic Decisions	
2024-11-25	The Imaging Wire	Fetal Ultrasound AI Cleared	
2024-12-02	The Imaging Wire	Milvue Joins deepc AI Platform	
2024-12-15	The Imaging Wire	Researchers Study AI for Pediatric Fractures	

Pediatric radiology AI in the trade press, 2025 (22 stories)

Date	Newsletter	Story	Paper
2025-01-08	The Imaging Wire	AI for Pediatric Fracture Detection	
2025-01-08	The Imaging Wire	Children's National taps Microsoft to prototype AI tools .	
2025-01-29	The Imaging Wire	Samsung Launches OB/GYN Ultrasound	
2025-02-02	The Imaging Wire	Ultrasound AI Detects Fetal Defects	
2025-02-06	ESR / ECR	Philips at #ECR2025: New AI-enabled systems, intelligent software, and...	
2025-02-14	ESR / ECR	Learn more about FUJIFILM's latest innovations at ECR!	Iwan et al., 2020 (European Radiology)
2025-02-27	The Imaging Wire	AI Detects Pediatric Epilepsy	
2025-03-04	TLDR AI	AI to diagnose invisible brain abnormalities in children with epilepsy...	
2025-04-02	The Imaging Wire	AZmed Gets New Chest AI Clearances	
2025-04-13	The Imaging Wire	GE's Cincinnati Pediatric Partnership	
2025-05-14	The Imaging Wire	Philips Partners with NVIDIA on AI for MRI	
2025-05-18	The Imaging Wire	New GE Clearances	
2025-05-20	RSNA News	Radiologists Share Tips to Prevent AI Bias	
2025-06-01	The Imaging Wire	BrightHeart's 3rd FDA Clearance	
2025-07-03	The Imaging Wire	DeepEcho Gets FDA Nod for Fetal AI	
2025-07-10	RSNA News	How Is AI Being Used in Daily Neuroradiology Practice?	
2025-08-03	The Imaging Wire	Radiologist Pay Jumps Nearly 8% in New Survey	
2025-08-13	The Imaging Wire	AI-Assisted Fracture Detection	
2025-08-27	The Imaging Wire	Sonio Launches New Maternal-Fetal AI Options	
2025-10-01	The Imaging Wire	Carebot Expands AI Portfolio	
2025-10-08	The Imaging Wire	Reducing CT Radiation Dose System-Wide	
2025-11-09	The Imaging Wire	AI Detects Pediatric Elbow Fractures	

Pediatric radiology AI in the trade press, 2026 (year to date) (13 stories)

Date	Newsletter	Story	Paper
2026-01-07	The Imaging Wire	BrightHeart Raises EUR 11M	
2026-01-28	The Imaging Wire	AHA Publishes New Stroke Guidelines	Prabhakaran et al., 2026 (Stroke)
2026-01-28	ESR / ECR	Guerbet: Innovation and French Cooperation in the Spotlight at ECR 2026	
2026-02-01	The Imaging Wire	DL-Based CT Reduces Radiation Dose	
2026-02-08	The Imaging Wire	GE Bolsters Ultrasound with Fetal AI	
2026-02-26	ESR / ECR	MRI Unveiling at the	
2026-03-29	The Imaging Wire	Ultrasound AI Gets Breakthrough Nod	
2026-05-10	The Imaging Wire	Pediatric MRI Safety, RTs and Remote Radiology, and PSMA-PET	
2026-06-22	RSNA News	Researchers Use AI to Establish Body Charts for CT Imaging Across the A...	
2026-06-28	The Imaging Wire	BrightHeart's Bright Future	Zelop et al., 2025 (Obstetrics & Gynecology)
2026-07-12	The Imaging Wire	AI's Rise in Pediatric Imaging	Shah et al., 2026 (European Radiology)
2026-08-16	The Imaging Wire	CT vs. MRI for Pediatric TBI, Helium Vulnerability, and Ditching the Di...	
2026-08-25	RSNA News	AI Challenges Fuel Innovation Beyond the Leaderboard	

Commercial software with a pediatric angle

Vendor	Product	Task	Pediatric status	FDA list
Gleamer	BoneView	fracture detection on radiographs	pediatric indication (age 2+) cleared 2023	4 (2022–2025)
AZmed	Rayvolve	fracture (and chest) findings on radiographs	pediatric fracture indication cleared 2024	4 (2022–2025)
Visiana	BoneXpert	automated bone age (GP / TW)	pediatric by design; CE-marked since 2009 (not on the FDA AI list)	not listed
16 Bit	Rho	automated bone age	pediatric by design	1 (2024)
Ever Fortune.AI	EFAI Bonesuite Bone Age Pro	automated bone age	pediatric by design	11 (2022–2026)
BrightHeart	Fetal EchoScan	fetal echocardiography view / anomaly assist	prenatal (fetal) by design	5 (2024–2025)
GE HealthCare / Canon / Siemens / Philips	TrueFidelity, AiCE, Deep Resolve, Precise Image	deep-learning CT / MR reconstruction (lower dose, faster scans)	adult-cleared scanner software, widely used for pediatric dose reduction	307 (2014–2026)
Subtle Medical / AIRS Medical	SubtleMR, SwiftMR	accelerated MRI via deep-learning denoising	adult-cleared; pediatric scan-time studies	13 (2018–2026)
Qure.ai	qXR / qER	chest radiograph and head CT triage / quantification	adult indications; pediatric TB-screening evidence only	9 (2020–2026)
Aidoc / Viz.ai	BriefCase, Viz LVO / ICH	worklist triage (hemorrhage, LVO, PE, pneumothorax)	adult-only indications; the largest deployed category	43 (2018–2026)

"FDA list" = number of radiology AI-enabled devices the company has on the FDA list (years). Pediatric status is the publicly stated indication; confirm the age range in the 510(k) summary before purchase.

Worth knowing as a radiologist — and why

- **TotalSegmentator** (open tool): free, one-command segmentation of 100+ structures on any CT (and MR); the basis for opportunistic measurements (bone density, muscle, organ volumes)
- **nnU-Net** (open tool): if a vendor claims a segmentation result, this is the baseline they had to beat
- **MONAI** (open framework): what your data scientists will build with; knowing the name lets you read a methods section
- **RSNA Pediatric Bone Age Challenge (2017)** (paper / dataset): the one pediatric task with a public benchmark, many models, and cleared products (BoneXpert, Rho)
- **"Variable generalization performance of a deep learning model to detect pneumonia in chest radiographs" (Zech et al., PLoS Med 2018)** (paper): the cleanest demonstration that a model can learn the hospital instead of the disease; the reason to demand local validation
- **CLAIM checklist (Radiology: AI 2020)** (guideline): how to tell a well-reported AI study from a weak one
- **"Improving Image Quality and Reducing Radiation Dose for Pediatric CT by Using Deep Learning Reconstruction" (Brady et al., Radiology 2021)** (paper): the most-cited non-bone-age pediatric paper: deep-learning reconstruction lowers pediatric CT dose with better image quality; the most immediately usable pediatric benefit
- **FDA AI-enabled device list** (registry): the authoritative list of what is cleared; check the indication's age range before buying
- **Gleamer BoneView / AZmed Rayvolve** (products): the first fracture-detection tools with pediatric indications; realistic first pediatric AI purchases
- **MedSAM / segment-anything models** (foundation models): where annotation and segmentation tooling is heading: click, don't draw

What the field does well

- **Finding what is there.** Detection / diagnosis is the bulk of the literature (45% of radiology-AI papers, 46% of pediatric); in adults it has become deployed worklist triage, the largest block of the 1,164 FDA-listed radiology AI devices.
- **Outlining and measuring.** Segmentation (26% of papers) has open, strong defaults (nnU-Net, TotalSegmentator); in children the mature measurement task is bone age (RSNA challenge: 4.2-month error, several cleared products), and measurement / quantification is 22% of pediatric papers.
- **Better images from less dose.** Reconstruction / imputation is 23% of radiology AI and 12% of pediatric work; deep-learning reconstruction is on today's scanners and cut pediatric CT dose by about half at equal or better image quality.
- **Shared benchmarks where they exist.** Fetal and neonatal brain MRI (FeTA, dHCP) and pediatric brain tumors (BraTS-PEDs) now have challenge datasets, and multi-site pediatric neuro-oncology models match radiologists retrospectively.

Bleeding edge

- **Foundation and vision-language models**: 7.2% of radiology-AI papers (3.9% pediatric); segment-anything style tools (MedSAM) and generalist radiology models are replacing task-specific training.
- **Report generation and LLMs**: 3.8% of papers; drafting and extracting from reports, with fluency ahead of reliability.
- **Agents**: 1.0% of radiology-AI papers (42 pediatric papers in total); LLMs that call imaging tools and take multi-step actions.
- **Pediatric-specific**: bone age robust to skeletal dysplasias (Deeplasia); fetal ultrasound AI is the most active commercial pediatric area in the trade press (BrightHeart, Sonio, DeepEcho clearances, 2023–2026); adult fracture tools extended to children (Gleamer 2023, AZmed 2024) and now tested in a real pediatric emergency department.
- **Generalization studies**: adult-trained CT organ segmentation degrades in small children, and the first pediatric prospective-style reader studies show smaller gains than stand-alone accuracy implies.

Open problems — where a children's hospital could contribute

- **Which diseases?** Since 2023 the pediatric corpus (6,723 papers) leans to autism / ADHD / neurodevelopment (697); fetal anomalies / prenatal (534); brain tumors (325); congenital heart disease / cardiac (295). Barely studied: testicular / ovarian torsion (8); inflammatory bowel disease (22); lines and tubes / NICU support devices (54); neonatal jaundice / biliary atresia / liver (56); craniosynostosis / craniofacial (72); appendicitis / intussusception / NEC (72). Which of these deserve a model first is an open question for this group.
- **Dataset curation.** What would a multi-center pediatric dataset look like: age-stratified, consented, protocol-diverse, with rare phenotypes? Which of our own archives (CT, ultrasound, NICU radiographs) could become the pediatric benchmark that does not yet exist?
- **Validation.** Local, age-stratified testing of adult-cleared tools; prospective reader studies rather than retrospective accuracy; monitoring for drift as children grow and protocols change.
- **Beyond interpretation.** Workflow / non-interpretive work is 13% of papers, but communication with families, education, protocoling and policy are almost absent; these are cheaper to build and evaluate than diagnostic models.
- **Regulation and liability.** Few of the 1,164 radiology clearances carry a pediatric indication; which adult clearances are acceptable to use off-label in children, and under what local validation?

Implications for a children's hospital

1. **Buy maturity, build for the gaps**: adopt cleared adult-derived tools that transfer (reconstruction, bone age, fracture with a pediatric indication); treat the under-studied pediatric problems as local validation and research.
2. **Demand local pediatric validation** before clinical use, by age group, and check the age range in the clearance.
3. **Prioritize dose and throughput**: deep-learning reconstruction gives the clearest pediatric benefit today.
4. **Invest in data**: a curated, shareable pediatric dataset is the scarcest resource in the field and the contribution a children's hospital is uniquely placed to make.